

Progress Report - 2026-04-21

Project: Image-Constrained Generative Modeling of Retinal Vasculature

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1. Current Scope

This stage focuses on a fundus-based implementation of the proposed image-constrained generative modeling framework. The original proposal includes both fundus and OCTA images, but OCTA data are not yet available. Therefore, the current model uses fundus images to extract structural constraints and treats OCTA integration as the next major extension.

The present results should be read as a proof-of-concept for the modeling pipeline rather than a full multimodal validation study.

2. Image Data Used

The local dataset contains 15 healthy fundus images:

01_h.jpg through 15_h.jpg

The visual figures use three representative images:

01_h.jpg, 05_h.jpg, and 10_h.jpg

These three examples are used for qualitative visualization because showing all 15 images would make the report crowded. Quantitative evaluation is run across all 15 available fundus images as a robustness check.

3. Implemented Pipeline

Fundus Constraint Extraction

For each fundus image, the pipeline extracts:

- retina field-of-view boundary
- optic disc location
- macula / foveal exclusion region
- major superior and inferior vessel orientations
- vessel segmentation mask

- regional vessel density map approximated from fundus vessel segmentation

The density map is currently fundus-derived. In the proposal, this role is ideally filled by OCTA because OCTA provides deeper microvascular and layer-specific density information.

Generative Models Compared

Three model variants are compared:

Model	Description
Baseline	Uses manually specified default branching parameters
Image-Constrained	Uses fundus-derived optic disc location and vessel orientation
Density-Aware	Adds regional density modulation to the constrained generator

4. Evaluation Design

The report separates visual demonstration from robustness evaluation:

Purpose	Images Used	Output
Qualitative visual comparison	3 representative fundus images	Constraint overlays and model comparison figures
Quantitative robustness check	All 15 available fundus images	CSV metrics and averaged summary chart

This avoids relying on a single attractive example while keeping the report readable.

The evaluation metrics include:

- number of edges
- number of terminal vessels
- total vessel length
- terminal coverage score
- fractal dimension estimate
- mean branch angle
- density correlation

5. Main Limitation

The current density-aware result is still limited because local density mainly affects branching depth rather than directly steering terminal placement. This is the main modeling weakness exposed by the current results.

The evaluation is also preliminary. The fractal-dimension calculation is rough, and a stronger topology- or distribution-based comparison would make the results more informative.

6. OCTA Status

OCTA remains part of the intended framework, but it is not yet included in the implemented experiments because the OCTA data have not been provided yet.

Once OCTA data become available, the next extension will be to move from the current fundus-only constraint model toward the multimodal framework originally proposed. OCTA would allow deeper vascular density, layer-specific microvascular structure, and the macula avascular zone to be incorporated directly rather than approximated only from fundus vessel segmentation.

7. Next Steps

1. Strengthen the density-aware branching rule so that local density affects terminal placement more directly.
2. Refine evaluation with stronger distribution- or topology-based metrics.
3. Integrate OCTA-derived density and FAZ constraints when OCTA data become available.
4. Continue using three representative images for clear figures while keeping all 15 fundus images in the batch evaluation.